DAVID KURNIAWAN

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EDUCATION

Xiamen University Malaysia | Selangor, Malaysia **Sept 2028**
BEng (Hons) in Artificial Intelligence **(GPA: 3.83 out of 4.00)**
Dean's List Awardee for 3 consecutive semesters — *College of Artificial Intelligence and Robotics • Top 16% of cohort*

TECHNICAL SKILLS

Languages: Python • HTML & CSS • JavaScript • C Programming • C++
ML / AI Frameworks: TensorFlow • Keras • Scikit-learn • Pandas • LangChain • OpenCV • PyTorch
Infra and Frameworks: FastAPI • REST API • ML-Agents • Docker • WSL • Flask • Node • React
Tools and Workflow: Git • Cursor • Antigravity • Claude Code • VS Code • PyCharm • Anaconda
Languages (Human): Indonesian (Native) • English (IELTS Band 7) • Chinese (Elementary)

SELECTED PROJECTS

KerjaCerdas | LangGraph, FastAPI, PostgreSQL (pgvector), Docker, React **2026**
Lead Engineer / Backend

- Architected an autonomous multi-agent recruitment swarm using **LangGraph** (Supervisor + parallel worker nodes) and **FastAPI** to solve labor market mismatch; automated candidate parsing, skill-gap analysis, and interview shortlisting via a unified orchestration layer.
- Engineered a 5-signal hybrid ranking engine in **PostgreSQL** (pgvector) combining 768-dim semantic vector search with structured candidate filters (skill overlap, salary fit, location) and **HNSW** indexing, achieving sub-200ms candidate retrieval latency.
- Built a **multi-modal CV processing pipeline** (PyMuPDF + Gemini Vision) with dynamic **Token Efficiency Gates** and **Regex input-sanitization middleware**, preventing prompt injection while **cutting LLM token overhead by ~40%.**
- Developed B2B monetization & enterprise infrastructure, including Pay-to-Unlock candidate teasers, simulated E-KYC verification, and a 5-stage automated **GitHub Actions CI/CD** pipeline with **live pgvector test containers.**

Orion | FastAPI, LangGraph, LangSmith, Pydantic, GitHub Actions, Python
2026
Tech Lead / Backend (4-person team)

- Built a 6-stage autonomous LLM workflow using **LangGraph** and **FastAPI** to automate end-to-end enterprise expense claim processing; parsed natural language submissions to extract, validate, and auto-approve claims, **reducing processing time by ~80%.**
- Integrated **LangSmith** observability & type-safe **Pydantic** contracts, establishing real-time tracing, token cost monitoring, and execution state debugging across all agent stages.
- Engineered deterministic document parsing & policy engines (Python, rapidfuzz, pypdf) for duplicate claim detection via **fuzzy string matching** and persistent audit logging.
- Established dual-tier CI/CD testing architecture (**GitHub Actions**): PR gates enforcing unit/integration tests at **85% code coverage**, alongside nightly cron regression suites running live against production LLM APIs.

NeuralVoid | Node.js, FastAPI, Scikit-Learn, Railway, Vercel **2026**

- Engineered an end-to-end behavioral analytics platform; built a **25-feature ML pipeline** extracting session length, binge streaks, and doomscroll velocity metrics from raw TikTok interaction logs.
- Trained a **soft-voting ensemble model** (XGBoost, Random Forest, Logistic Regression) achieving **~96% accuracy** in detecting compulsive usage patterns.
- Integrated Google Gemini AI via **FastAPI** backend to auto-generate clinical behavioral reports from real-time model outputs.
- Developed a responsive **React** analytics dashboard with interactive heatmaps and radar charts, deployed across **Vercel** (Frontend) and **Railway** (Backend API).

AWARDS AND CERTIFICATIONS

Silver Award — SEA-CICSIC 2026 **June 2026**
China-ASEAN Innovation Competition (Undergraduate Category)

- Led AI strategy and technical architecture for Omni-QC, a manufacturing intelligence platform featuring real-time defect prediction and predictive quality control, competing at the China-ASEAN undergraduate level.
- Designed the data-driven predictive system architecture underpinning real-time defect detection and operational decision-making for industrial manufacturing environments.

3rd Place — DPickleball AI Competitions (Unity • ML-Agents • Reinforcement Learning) **Oct 2025**
XMUM AI Club • Xiamen University Malaysia

- Served as lead developer of 3-member team, architected the full training system design including environment setup via Gymnasium, custom reward shaping strategy, PPO algorithm implementation, and end-to-end training pipeline, directing technical execution across the competition team.
- Engineered the reward shaping function and PPO (Proximal Policy Optimisation) training loop using Unity ML-Agents and Gymnasium, iteratively tuning hyperparameters to achieve stable agent convergence and competitive autonomous play in digital pickleball environment.

Top 20% Globally — International Quant Championship, Stage 1 **Apr 2025**

World Quant BRAIN

- Placed top 20% globally in an international quantitative reasoning competition, benchmarking analytical and data-driven problem-solving skills against international competitors.

RESEARCH INTEREST

Artificial Intelligence • Agentic AI • Deep Learning • Quantitative Modelling • Robotic Learning • Sim-to-Real Reinforcement Learning • Autonomous Systems

LEADERSHIP & ORGANISATION EXPERIENCE

Student Council — Tritunggal Christian High School **Oct 2021 - Oct 2023**

Head of Treasury Department

- Managed and reported school event finances across 10+ annual events including class meetings, Teacher’s Day, Independence Day, and orientation, maintaining full financial accountability.
- Led a fundraising campaign through canteen sales initiatives, generating 4 million rupiah in revenue, demonstrating budget planning and cross-team coordination skills.